Analysis of Household Surveys

A five-session workshop

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Department of Economics, University of Ghana, hosted by the International Growth Centre. Five meetings of two hours, 11am–1pm, in CCC.

1 Administrivia

1.1 Meetings

Session	Day	Date	Topic
1	Tuesday	8 September	Design, sampling, and collection
2	Thursday	10 September	Describing welfare
3	Monday	14 September	Demand and welfare
4	Tuesday	15 September	From cross-sections to dynamics
5	Thursday	17 September	Capstone and frontiers

1.2 Texts

Angus Deaton's *The Analysis of Household Surveys: A Microeconometric Approach to Development Policy* (Johns Hopkins, 1997) is the spine of the workshop. The World Bank's 2018 reissue is open access, so it costs you nothing; the workshop page carries the link. Chapter 1 repays reading before the first session.

Two supplements, each for a particular session:

- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates for Welfare Analysis* (World Bank LSMS Working Paper 135, 2002). Session 2.

- Deaton (1985), "Panel data from time series of cross-sections," *Journal of Econometrics* 30:109–126. Session 4.

1.3 Software

The hands-on work runs on a shared JupyterHub at hhsurveys.ligonresearch.org. You will be given a username and set your own password at first login (be sure to jot this down!). Nothing *must* to be installed to take part, but I encourage you to also install the relevant software on your own computer (this will be necessary to use these tools after the workshop). The workshop page has instructions.

The computation uses `LSMS_Library`, which gives one interface to the household surveys of some forty countries. Installing it on your own machine is worth doing. Do it at home: it is a 1.3 GB download, and twenty laptops fetching it at once over classroom wifi will simply stall.

1.4 What you should be able to do afterwards

Not "know about," but "do." By the end of the fifth session you should be able to take an unfamiliar LSMS-style survey and:

1. Read its documentation well enough to say what population it represents, how it was drawn, and which of its variables the design lets you use;

2. Construct a defensible consumption aggregate from its modules, deflate it across space and time, and put an equivalence scale on it;

3. Produce weighted, correctly-clustered estimates of poverty and inequality, with standard errors you are willing to defend;

4. Estimate Engel curves and a simple demand system, and say what they imply for the welfare consequences of a price change;

5. Build a pseudo-panel from repeated cross-sections, and know when to prefer it to a real panel and when not;

6. Hand the whole thing to a colleague as code that runs.

1.5 Assessment

There is no assessment: this is a workshop, not a course. The fifth session is given over to a small analysis of a survey of your own choosing, which you will present to the room in about five minutes. Come to session 4 with a candidate country in mind.

2 Sessions

2.1 Design, Sampling, and Collection

What a careful analyst should know before touching the data.

2.1.1 Reading

- Deaton ch. 1 — the whole chapter
- Deaton §1.4 on survey design and §2.2 on what to do about it
- Skim the GLSS7 *Report* methodology section

Deaton is open access (CC BY 3.0 IGO). On the hub it is already in your `reading/` folder; otherwise PDF.

The GLSS7 report is published by the Ghana Statistical Service and is not ours to redistribute: GSS catalogue 97.

2.1.2 Brief description of LSMS Surveys

A timeline of the surveys `lsms_library` knows about: one row per country, one bar per survey wave, so we can see overall temporal coverage — 37 countries, 113 waves, 1985 to the present.

The same page carries a graphical version of the coverage matrix: which features the library grades, wave by wave, and how complete each one is.

`https://claude.ai/code/artifact/7746f698-7b43-49d0-b2fc-22f7a1fb6496`
(Also linked from the workshop page as *LSMS Library Coverage*.)

2.2 Describing Welfare

Constructing a consumption aggregate, and then describing the distribution of it, before modelling anything.

2.2.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton, chs. 3 and 4
- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates* — the operational reference, and the one you will actually reach for
- Deaton & Zaidi §§2–4 if you read nothing else

2.3 Demand and Welfare

Where household surveys earn their keep for policy: what happens to whom when a price moves.

2.3.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton, chs. 4 (§4.2 on Engel curves) and 5 (all)
- Deaton & Muellbauer (1980), "An Almost Ideal Demand System," *AER* 70:312–326
- Lewbel (1991), "The rank of demand systems," *Econometrica* 59:711–730
- Ligon, "Household Welfare from Disaggregate Demands" (working paper) — the CFE system we estimate today

2.4 From Cross-Sections to Dynamics

Two ways to get change out of survey data: build a panel from cohorts, or use one that already exists.

2.4.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton (1985), "Panel data from time series of cross-sections"
- Deaton, ch. 2 §2.7 and ch. 6
- Abadie, Athey, Imbens & Wooldridge (2023), "When should you adjust standard errors for clustering?" *QJE* 138:1–35

2.5 Capstone and Frontiers

Where the methods break down, and then your own analysis.

2.5.1 Reading